CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 15: Linear Inequations

Topic-wise short notes with solved examples

Linear inequations in one variable, the rules for solving them, the replacement set and the solution set, and showing the solution on the number line.

1 Inequations

- An **inequation** compares two expressions with one of the signs $<$, $>$, $\leq$, $\geq$.

Sign	Read as
$<$	is less than
$>$	is greater than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to

- A **linear inequation in one variable** has the form $ax + b < c$ (or with $>$, $\leq$, $\geq$), where $a \neq 0$.
- $x < 5$ and $5 > x$ mean the same thing.

2 Replacement set and solution set

- The **replacement set** (domain) is the set from which values of the variable are chosen. It is given in the question.
- The **solution set** is the set of all values from the replacement set that satisfy the inequation.
- The same inequation can have different solution sets for different replacement sets.

Symbol	Set	Elements
N	Natural numbers	$1, 2, 3, \ldots$
W	Whole numbers	$0, 1, 2, 3, \ldots$
Z or I	Integers	$\ldots, -2, -1, 0, 1, 2, \ldots$
R	Real numbers	All points on the number line

Example 2.1 Write the solution set of $x < 4$ when the replacement set is (i) N (ii) W (iii) $\{-2, 0, 3, 5, 6\}$.

SOLUTION

(i) $\{1, 2, 3\}$ **(ii)** $\{0, 1, 2, 3\}$ **(iii)** $\{-2, 0, 3\}$

3 Rules for solving

- **Adding or subtracting** the same number on both sides does **not** change the sign.
- **Multiplying or dividing** both sides by the same **positive** number does **not** change the sign.
- **Multiplying or dividing** both sides by the same **negative** number **reverses** the sign: $<$ becomes $>$ and $\leq$ becomes $\geq$.
- A term may be transposed from one side to the other by changing its sign, just as in an equation.
- If the sides are interchanged, the sign reverses: $a > b$ is the same as $b < a$.

Example 3.1 Solve $3x - 5 < 10$, $x \in N$.

SOLUTION

$3x < 15$, so $x < 5$

$\{1, 2, 3, 4\}$

Example 3.2 Solve $-2x \geq 6$, $x \in Z$.

SOLUTION

Divide by -2 and reverse the sign: $x \leq -3$.

$\{\ldots, -5, -4, -3\}$

Example 3.3 Solve $7 - 3x > 1$, $x \in W$.

SOLUTION

$-3x > -6$, so $x < 2$

$\{0, 1\}$

Example 3.4 Solve $5x + 2 \leq 3x + 12$, $x \in N$.

SOLUTION

$2x \leq 10$, so $x \leq 5$

$\{1, 2, 3, 4, 5\}$

Example 3.5 Solve $\frac{2x - 1}{3} > 3$, $x \in \{1, 2, 3, \ldots, 8\}$.

SOLUTION

$2x - 1 > 9$, so $x > 5$

$\{6, 7, 8\}$

Example 3.6 Solve $\frac{x}{2} - \frac{x}{3} \leq 1, x \in W$.

SOLUTION

Multiply by 6: $3x - 2x \leq 6$

$\{0, 1, 2, 3, 4, 5, 6\}$

4 On the number line

- For a solution set of **integers, whole or natural numbers**: mark each solution with a thick dot. Three dots beyond the last marked number show that the set continues.
- For **real numbers**: draw a thick line over the part that is included. Use a **solid dot** at an end point that is included ($\leq$ or $\geq$) and a **hollow circle** at an end point that is not included ($<$ or $>$).



Solution set {−3, −2, −1, 0} of −3 ≤ x < 1, x ∈ Z, shown by dots



−2 ≤ x < 2, x ∈ R: solid dot at −2 (included), hollow circle at 2 (not included)

5 Combined inequations

- A statement such as $-3 < 2x - 1 \leq 5$ contains two inequations. Solve by doing the same operation to **all three parts**.
- Or split it into two inequations, solve each, and take the values common to both.

Example 5.1 Solve $-3 < 2x - 1 \leq 5, x \in Z$.

SOLUTION

Add 1: $-2 < 2x \leq 6$. Divide by 2: $-1 < x \leq 3$.

$\{0, 1, 2, 3\}$

Example 5.2 Solve $-5 \leq 3x + 1 < 10, x \in R$, and describe the graph.

SOLUTION

Subtract 1: $-6 \leq 3x < 9$. Divide by 3: $-2 \leq x < 3$.

All real numbers from -2 to 3; solid dot at -2, hollow circle at 3

Example 5.3 Solve $2 \leq 5 - x < 6, x \in Z$.

SOLUTION

Subtract 5: $-3 \leq -x < 1$. Multiply by -1 and reverse: $3 \geq x > -1$.

$\{0, 1, 2, 3\}$

6 Greatest and least values

Example 6.1 Find the greatest value of x satisfying $4x - 3 \leq 17, x \in N$, and the least value of x satisfying $2x + 1 > 9, x \in N$.

SOLUTION

$x \leq 5$; and $x > 4$

Greatest 5; least 5

Example 6.2 Find the smallest integer satisfying $3x + 8 > 2$.

SOLUTION

$3x > -6$, so $x > -2$

-1

7 Common mistakes

- Not reversing the sign when multiplying or dividing by a negative number.
- Reversing the sign when adding or subtracting, or when dividing by a positive number.
- Ignoring the replacement set and listing numbers that are not in it.
- Including an end point when the sign is $<$ or $>$.
- Using a solid dot where a hollow circle is needed.
- Listing 0 as a natural number.